AURA

Autonomous User Realization & Assembly

A Cyber-Physical Inventory-Aware Electronics Assistance System for Distributed Component Localization and Deterministic Circuit Composition

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Abstract - AURA addresses a practical electronics bottleneck: users may know what they want to build, yet still lose time checking whether the exact parts are available and physically searching storage to find them. The platform combines an ESP32 host, low-power ATtiny404 plus nRF24L01 locator nodes, WS2812-based indication, quantity-aware inventory metadata, and a deterministic JSON-first circuit workspace. Within the workspace, local calculations and topology recognition ground AI-assisted circuit creation, explanation, and revision so that the design state remains auditable instead of becoming unstructured chat output. AURA therefore links part search, value-level stock checking, structured circuit authoring, and physical retrieval of required components. The current prototype supports single-part lookup, inventory-constrained circuit proposals, whole-project highlighting of all required parts, and an illustrative preview layer that is clearly labeled as not electrically verified.

Index Terms - cyber-physical systems, electronics inventory, component localization, inventory-aware circuit generation, deterministic circuit authoring, distributed node networks

AURA should be understood from the first page as a linked workflow: a locator network for physical search, value-aware inventory records, an AI-assisted circuit creator constrained by local deterministic logic, a canonical JSON workspace for review and revision, and an honest preview layer that improves intuition without claiming verified simulation.

TABLE I. AURA AT A GLANCE

Element	Function in AURA	Immediate Benefit
Locator network	Addressed host-to-node commands illuminate exact storage positions for requested parts or whole project sets	Reduces drawer-by-drawer search and makes the room itself part of the interface
Inventory intelligence	Tracks family, exact value, quantity, and location instead of only broad categories	Grounds feasibility in the parts the user actually owns
AI-assisted circuit creator	Returns canonical circuit JSON or structured patches after inventory filtering and local grounding	Helps users start or revise circuits without losing structure and reviewability
Deterministic workspace	Keeps explicit components, anchors, wires, analysis, preview, and compare flow aligned around canonical JSON	Improves trust because visible state remains auditable
Honest preview layer	Shows illustrative behavior while explicitly labeling it as not electrically verified	Improves understanding without overstating confidence

I. INTRODUCTION

Electronics prototyping often breaks down between idea and assembly. A user may know the circuit goal, yet still spend disproportionate time checking whether the exact resistor values, switches, transistors, or timers are available and then manually searching storage to locate them.

Most existing tools address only one fragment of this process. Simulator-first applications assume parts exist. Inventory applications may preserve records but rarely illuminate physical locations. General-purpose AI can propose circuits, but often ignores owned stock, value-level variants, and the practical step of retrieving the exact parts.

AURA combines a cyber-physical locator system for real components and a deterministic circuit workspace for creating, explaining, and revising buildable designs. The goal is to reduce the total time and uncertainty between wanting a circuit and physically gathering the parts needed to assemble it.

A. Practical Problems That AURA Targets

- **Location uncertainty:** Users often know that a component exists somewhere in storage but do not know which drawer, strip, or box actually contains it.
- **Variant uncertainty:** Owning a component family is not enough; build feasibility depends on exact values such as 220-ohm, 1-kilo-ohm, or 10-kilo-ohm resistors being available.
- **Quantity uncertainty:** Even when the correct value exists, the available count may still be insufficient for the intended project.
- **Workflow fragmentation:** Circuit recommendation, stock checking, explanation, and physical retrieval usually happen in separate tools or entirely manual steps.
- **Project retrieval inefficiency:** Users commonly gather parts one-by-one instead of receiving a room-scale overview.

TABLE II. PRACTICAL WORKFLOW FAILURES AND AURA RESPONSES

Problem	Conventional Outcome	AURA Response
Hidden component locations	Manual drawer search and label reading	Addressed locator nodes illuminate the real storage position immediately
Unknown value-level availability	Users know a category exists but not whether the correct value exists	Inventory records track family, variant, and count rather than category alone
Circuit suggestions ignore stock	Recommended circuits may require unavailable parts	Circuit proposals are constrained by inventory-aware reasoning and structured review
High-friction retrieval	Parts are gathered serially with repeated room traversal	Whole-project highlighting reveals all required component locations together
Disconnected hardware and software	Digital planning and physical retrieval remain separate tasks	AURA links query, review, and locate actions inside one workflow

B. What Makes AURA Different

- **Cyber-physical loop:** The system does not stop at recommending a part or a circuit; it can translate accepted results into real addressed locate actions in the storage environment.
- **Inventory-first feasibility:** AURA distinguishes between abstract component categories and exact owned values, quantities, and storage addresses.

- **Deterministic AI boundary:** AI assists with generation, explanation, and revision, but canonical circuit JSON and local calculations remain the trusted substrate.
- **Project-scale retrieval:** The system supports not only single-part lookup, but also simultaneous highlighting of all components required for a chosen design.

II. SYSTEM ARCHITECTURE

AURA is organized into four tightly coupled layers: a host controller and bridge layer, a distributed node layer, an inventory and metadata layer, and a deterministic circuit assistance layer.

TABLE III. AURA PROTOTYPE SUBSYSTEM SUMMARY

Subsystem	Implementation Focus	Operational Role
Host controller	ESP32, 1.8-inch display, joystick, Bluetooth bridge	Local search, metadata storage, mobile mediation, node command dispatch
Locator node	ATtiny404, nRF24L01, WS2812 outputs, optional buzzer	Addressed physical highlighting for drawers, racks, or project markers
Inventory layer	Family, value, quantity, address, media tags	Variant-aware availability checks and retrieval mapping
Circuit workspace	Canonical JSON, local analysis, structured AI proposals	Deterministic authoring, comparison, and locate handoff

A. Host Controller and Bridge Layer

The ESP32-based host provides the control plane, local search interface, address lookup, and Bluetooth mediation. A phone may act as a richer search front end, but the host remains the authority for hardware-level node communication.

Fig. 1. Current AURA host prototype used to validate local search and addressed node activation.

B. Distributed Node Layer

Compact ATtiny404 plus nRF24L01 nodes drive WS2812-based visual indication and optional audible cues. Their reach can be adapted to dense strips of drawers or more sparse project-storage layouts, allowing cost to follow spatial density.

Fig. 2. Distributed AURA node prototype illustrating the low-power addressed indication approach.

C. Deterministic Circuit Assistance Layer

- **Canonical JSON model:** Keeps circuit data explicit, serializable, and reviewable instead of letting the visible layout become the source of truth.
- **Local deterministic analysis:** Handles recognized patterns and calculations before any probabilistic AI suggestion is accepted.
- **AI-assisted circuit creator:** Returns deterministic circuit JSON or structured patches so design changes can be compared, applied, or rejected deliberately.
- **Honest preview boundary:** Preview behavior improves explanation and intuition, but it is explicitly not presented as fully verified electrical simulation.

TABLE IV. CURRENT SOFTWARE REALIZATION STACK

Tool or Layer	Current Use in AURA	Why It Matters
Vite UI runtime	Hosts the web-based deterministic workspace	Supports responsive local-first editing during circuit authoring
Fastify backend	Provides circuit, inventory, and proposal endpoints	Keeps persistence and backend actions separate from immediate canvas interaction
better-sqlite3	Stores durable local data	Fits the local-first workflow without unnecessary infrastructure
Ajv validation	Validates structured JSON	Protects the deterministic model from malformed state
fast-json-patch	Supports structured proposal comparison and application	Makes AI-assisted revision auditable instead of free-form
Vitest	Covers prototype-level verification	Strengthens confidence that the JSON-first workflow is stable and reproducible

III. USER WORKFLOW AND WORKSPACE INTERPRETATION

Readers usually understand AURA fastest through the user workflow. The interface is not decorative packaging around an AI feature. Each visible surface corresponds to inventory fit, canonical circuit state, analysis, preview, and physical locate handoff.

TABLE V. END-TO-END AURA USER WORKFLOW

Stage	System Action	User-Facing Outcome
Goal or query capture	User searches a part or describes a circuit objective	The request becomes a practical search or design task
Inventory fit check	Family, exact value, quantity, and location are checked	Build feasibility is grounded in owned parts rather than assumption
Deterministic circuit creation	Templates, local rules, and AI proposals produce canonical JSON or structured patches	The circuit remains inspectable, editable, and comparable
Analysis and explanation	Known patterns, calculations, warnings, and assumptions are surfaced locally	The user sees grounded reasoning instead of unstructured suggestion text alone
Preview behavior	Supported outputs are visualized with explicit trust labels	The user gains intuition while still seeing that the preview is not electrically verified
Locate and gather	Accepted parts are mapped to addressed nodes	The recommendation becomes immediately actionable in the real room

Fig. 3. AURA workspace prototype showing deterministic circuit review, inventory context, and trust-labeled explanation surfaces.

A. Workspace Modes and What They Mean

- **Inventory mode:** Focuses on owned parts, search readiness, and the component review path that prepares new assets before they become supported AURA parts.
- **Workspace mode:** Centers the canonical circuit canvas while keeping assistant, analysis, preview, and JSON views aligned around the same source of truth.
- **Compare mode:** Presents structured proposal review so the user can inspect what AI changed before accepting or rejecting a revision.

B. Reading the Interface from Left to Right

The left-side surface answers the feasibility question early through owned-part browsing, templates, and inventory pressure; the center canvas keeps components, anchors, wires, coordinates, and values explicit; the right-side surface concentrates the assistant, inspector, analysis, preview, and JSON detail.

Fig. 4. Left-side inventory surface used for owned-part browsing, template starting points, and early feasibility framing.

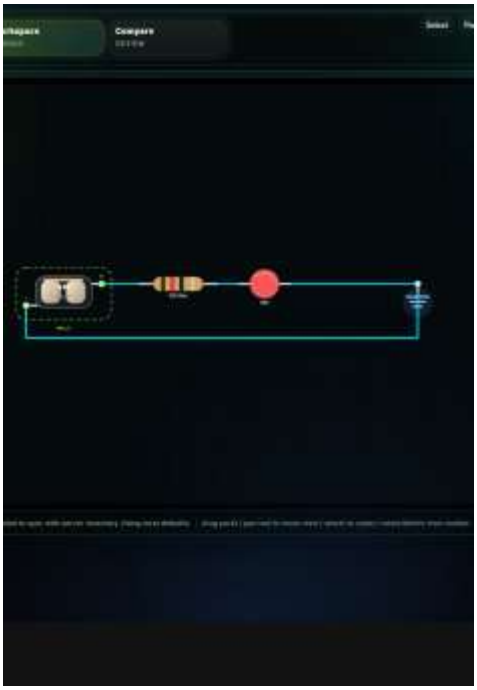

Fig. 5. Center circuit surface showing explicit components, connections, and the deterministic canvas structure.

Fig. 6. Right-side assistant, inspector, and detail surface showing how explanation and preview stay tied to structured circuit state.

- **Immediate feasibility:** Inventory status is visible early, so the user can see whether a recommendation is realistically buildable.
- **Review before action:** Structured compare and patch handling keep the user in control of AI-assisted circuit changes.
- **Honest learning support:** Local analysis and preview help the user understand likely behavior without pretending that unsupported circuits have been electrically proven.
- **Physical closure of the loop:** Accepted circuits can transition into locate actions, which gives the design process a concrete real-world endpoint.

IV. PROTOTYPE IMPLEMENTATION, BENEFITS, AND OUTLOOK

The present implementation is a complete prototype-level realization of the proposed workflow. The major subsystems are integrated strongly enough to demonstrate inventory-aware search, structured circuit recommendation and review, preview behavior framing, and physical part localization through addressed nodes.

A. Implemented Operating Modes

- **Single-component lookup:** The user searches for a part name or value and receives immediate physical highlighting of the relevant location.
- **Inventory-constrained circuit recommendation:** Circuit suggestions are filtered through owned parts and value-aware stock data rather than assuming infinite availability.
- **Whole-project highlighting:** After a circuit or parts list is accepted, AURA can illuminate all required component positions together instead of forcing serial retrieval.
- **Project marker mode:** Indicator nodes can be placed near completed hardware projects so retrieval extends beyond loose components and into project storage itself.
- **Inventory studio and component intake:** Legacy manifests and SVG assets can be reviewed in a deterministic workspace before a future compiler path turns them into native AURA components.

B. Representative Benefits

- **Students and beginners:** Spend less time on drawer-by-drawer search and more time understanding how a proposed design maps to real components.
- **Hobbyists and makers:** Can request practical circuits from the parts they already own rather than starting from idealized designs they cannot assemble.
- **Small labs and shared rooms:** Gain clearer inventory discipline and faster retrieval without needing enterprise-scale stock management infrastructure.
- **Project maintenance workflows:** Can identify and revisit stored builds more easily when project-linked indicators and metadata are present.

TABLE VI. PROTOTYPE VALIDATION SCENARIOS AND OBSERVED OUTCOMES

Validation Scenario	Expected Behavior	Observed Outcome
Single-item lookup	One requested component resolves to one addressed indicator	Targeted host-to-node activation demonstrated in controlled prototype operation
Dense-storage highlighting	One node covers multiple nearby locations economically	Architecture supports multi-position WS2812 indication for drawer strips and aligned bins
Value-aware stock check	Variant and quantity are checked before circuit use	Inventory model preserves value-level and count-level records for parts such as resistors
Circuit-to-locate handoff	Accepted circuit becomes an actionable retrieval set	Workspace flow preserves structured parts data for immediate localization
Whole-project retrieval	Multiple required components illuminate together	Grouped address mapping supports simultaneous project-level highlighting

C. Discussion and Conclusion

AURA should not be evaluated as a claim of universal circuit correctness or full simulator coverage. Its value lies in unifying physical inventory, quantity awareness, structured circuit composition, explanation, and retrieval actions within a single cyber-physical workflow.

The current prototype already proves an integrated architecture, prototype-level completeness, a practical inventory scope, and a clear trust boundary in which preview behavior is illustrative rather than electrically verified.

Future work will focus on PCB refinement, larger node deployments, richer smartphone interaction, and broader component coverage without abandoning the deterministic foundation that makes the system inspectable and trustworthy.

In summary, AURA presents a practical alternative to both simulator-only tools and unconstrained AI suggestions: a system in which digital recommendations remain auditable, inventory-aware, and immediately actionable in physical space.

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